THE NOVA RESEARCH CURRICULUM

General Science and Social Studies

Grade 3 · Term 2

Student edition

General Science: SC-3.1 life cycles, SC-3.4 energy sources and shadows. Social Studies: SS-3.3 natural resources and conservation, SS-3.4 inventions.

General Science · lesson notes

SC-3.1 Life cycles (Weeks 1–4)

Seed → seedling → plant → flower → seed. Egg → chick → hen. Egg → caterpillar → pupa → butterfly. Grow a bean in a jar and record its stages.

SC-3.4 Energy and shadows (Weeks 5–12)

Energy sources: sun, wind, water, food, fuel (gas, petrol), electricity. Shadows form when an object blocks light. Track a stick's shadow at 9 a.m., 12 noon and 3 p.m.: it is shortest at noon and points away from the sun.

Social Studies · lesson notes

SS-3.3 Natural resources and conservation (Weeks 1–8)

Resources: water, soil, forests, minerals, natural gas. Pakistan examples: the Indus River for farming, Sui gas, Khewra salt mine. Conserve by saving water and electricity, planting trees, recycling.

SS-3.4 Inventions (Weeks 9–12)

The wheel, printing, electric light, the telephone, the computer, the mobile phone. Compare life then and now; ask a grandparent what has changed.

▶ **Try it on NovaSchool** (open **Simulations** on your dashboard):

- **Life cycles** — Pick a living thing and step through its life cycle.
- **Shadows through the day** — Move the Sun across the sky from morning to evening. Watch the stick's shadow.

Workbook pages

Page	Outcome	Activity
1–2	SC-3.1	Draw and label two life cycles; bean diary.
3–4	SC-3.4	Energy sources sort; shadow record sheet.
5–6	SS-3.3	Resources map of Pakistan; my saving plan.
7	SS-3.4	Then and now: interview a grandparent.

Question bank

#	Outcome	Question	Your answer
1	SC-3.1	A caterpillar becomes a …	bird / butterfly / frog / bee
2	SC-3.1	A chick comes from an egg. True or false.	True / False
3	SC-3.1	Put in order: plant, seed, seedling.	_____
4	SC-3.4	Which is a source of energy?	the sun / a stone / a chair / sand
5	SC-3.4	A shadow forms when an object blocks light. True or false.	True / False
6	SC-3.4	When is a shadow shortest outdoors?	morning / noon / evening / night
7	SC-3.4	Food gives our body energy. True or false.	True / False
8	SC-3.4	Why do shadows change during the day?	_____
9	SS-3.3	Which is a natural resource?	plastic bag / water / car / phone
10	SS-3.3	The Khewra mine in Pakistan produces …	gold / salt / oil / sugar
11	SS-3.3	Planting trees helps conserve resources. True or false.	True / False
12	SS-3.4	Which invention lets us talk to people far away?	the wheel / the telephone / printing / the plough
13	SS-3.4	Write one way your life is different from your grandparents' childhood.	_____